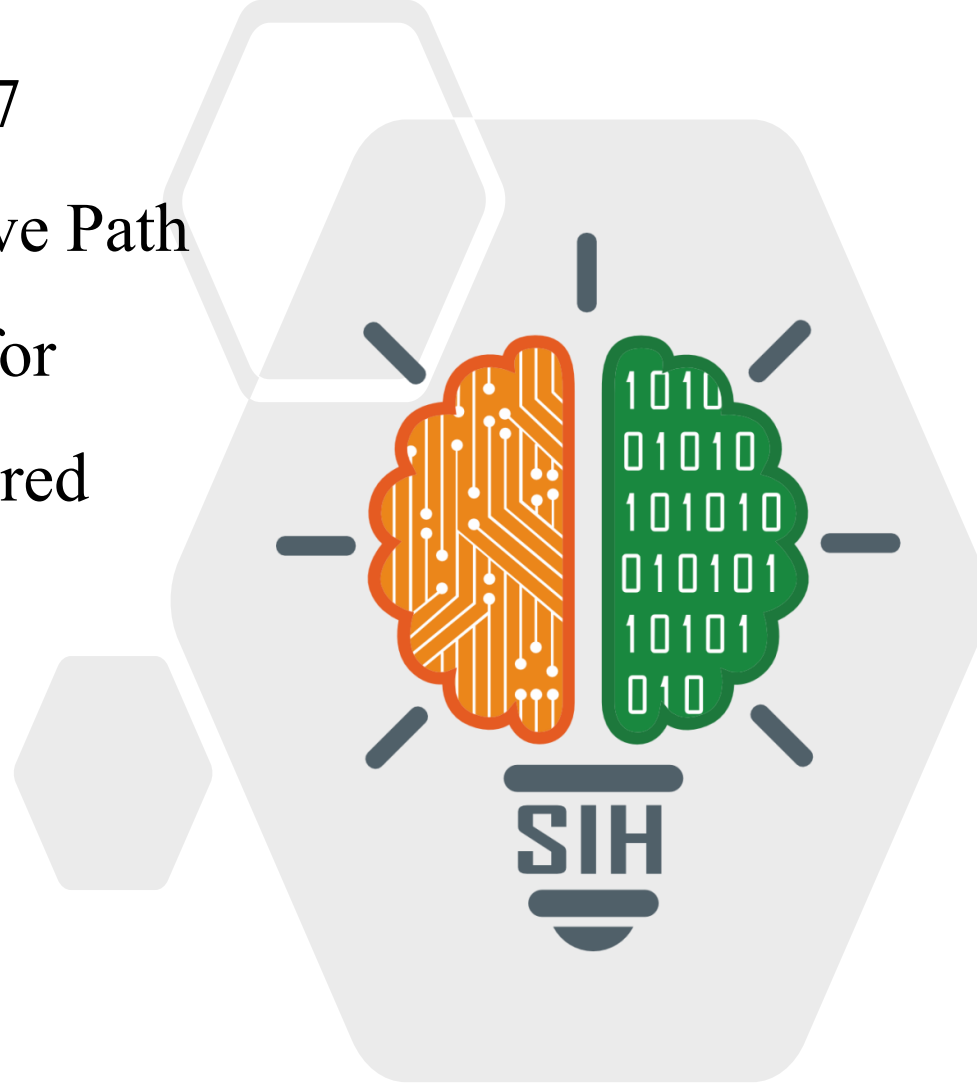


SMART INDIA HACKATHON 2026



- **Problem Statement ID** - SIH26037
- **Problem Statement Title** - Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme** - Smart Vehicles
- **PS Category** - Software
- **Team ID** - 174313
- **Team Name** - LIFIONYX



➤ Proposed Solution

Our closed-loop system combines multi-sensor perception, multi-hypothesis motion prediction, risk-aware behaviour planning and real-time trajectory replanning for unstructured Indian roads. It handles missing lane markings, informal merges, pedestrians, potholes, animals and mixed traffic, continuously adapting the vehicle path as road conditions change.

➤ How it addresses the problem

- ❖ Handles **missing or weak lane markings** and open drivable space.
- ❖ Detects and tracks **vehicles, pedestrians, animals** and static obstacles.
- ❖ Predicts **irregular and non-lane-based road-user motion**.
- ❖ Replans when **risk, obstacle position** or traffic behavior changes.
- ❖ Validates decisions through **repeatable closed-loop simulation**.

➤ Innovation and uniqueness of the solution

- ❖ Multi-Hypothesis Prediction
- ❖ Dynamic Risk Representation
- ❖ India-Specific Behavior Logic
- ❖ Closed-Loop Repeatable Validation

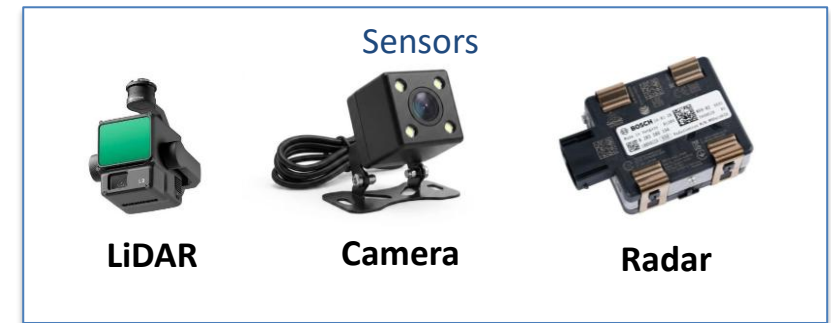
Why we stand out?

Scenario-Driven, Not generic Demo-Driven

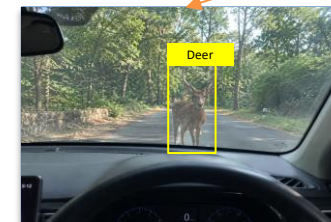
We Plan for Possibilities, Not One Prediction

End-to-End Closed-Loop Validation

Five Scenarios Are Only Our Starting Point

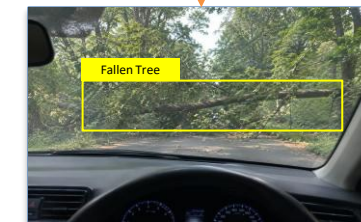


YOLOX-S Object Detection



Safe Animal

Wait



Static Obstacle

Replan Immediate



Unsafe Animal

Replan Immediate

Core Tech Stack

PERCEPTION

- ❖ RGB Camera
- ❖ LiDAR
- ❖ Radar
- ❖ Sensor Time Synchronization
- ❖ Calibration & preprocessing

OBJECT DETECTION

- ❖ YOLOX-S
- ❖ Anchor-free detector
- ❖ Decoupled head
- ❖ SimOTA label assignment

TRACKING & FUSION

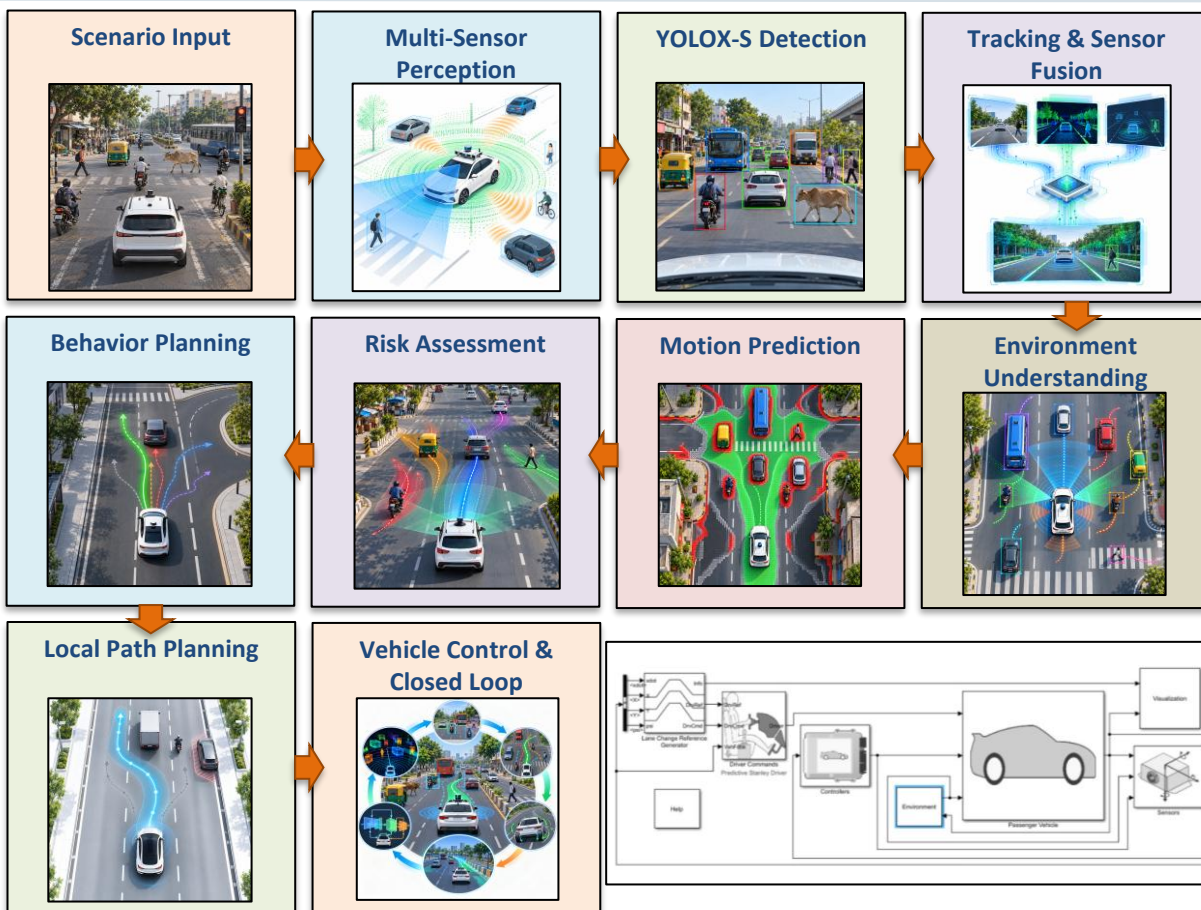
- ❖ Multi-object tracking
- ❖ Hybrid A*
- ❖ Radar-LiDAR-camera fusion
- ❖ Dynamic obstacle state update

PLANNING & CONTROL

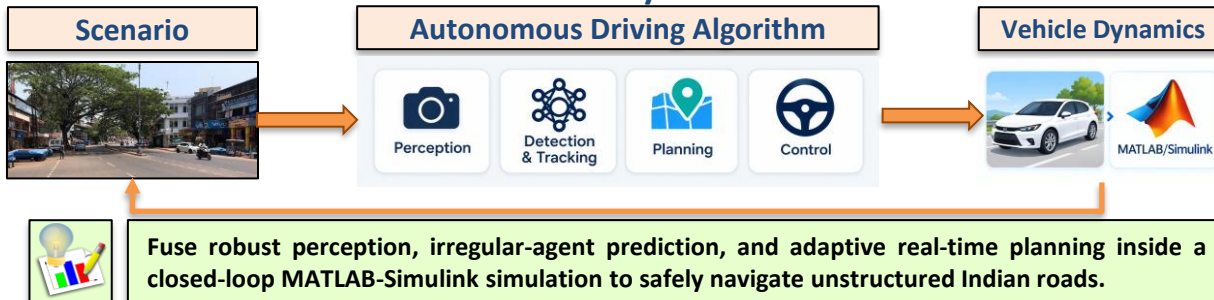
- ❖ Occupancy / cost map
- ❖ Behavior planning with Stateflow
- ❖ Path tracking controller

SIMULATION & VALIDATION

- ❖ MATLAB / Simulink
- ❖ Automated Driving Toolbox
- ❖ Navigation Toolbox
- ❖ Vehicle Dynamics Blockset
- ❖ Deep Learning Toolbox



CLOSED-LOOP MATLAB / SIMULINK VALIDATION

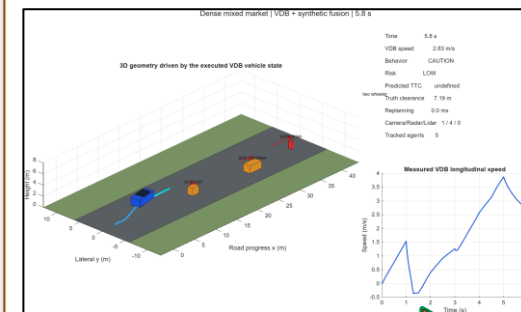


WHY YOLOX-S?

- ✓ Lightweight and fast for real-time inference
- ✓ Strong accuracy-speed trade-off
- ✓ Suitable for transfer learning on Indian mixed-traffic classes
- ❖ AP: 40.5
- ❖ Parameters: 9.0M
- ❖ FLOPs: 26.8G
- ❖ Speed: 9.8 ms on V100

Metrics

- ✓ Replanning latency
- ✓ Path smoothness
- ✓ Collision-free navigation
- ✓ Minimum clearance / near-miss safety margin



Simulation Result

Our Works



Why the Idea is Feasible

- ❖ **Closed-loop proof exists**
MATLAB/Simulink benchmark executes all 5 Indian-road scenario families.
- ❖ **India-specific data exists**
IDD supplies detection, segmentation, temporal + multimodal/LiDAR data.
- ❖ **Custom work is bounded**
Build risk trigger, motion hypotheses, trajectory cost + fail-safe

Why it is Viable

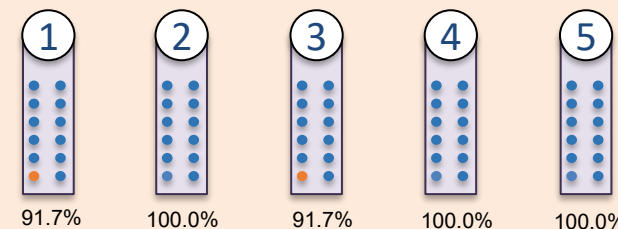
- ❖ **Software-first Pilot**
Simulation-first proof: repeatable experiments without vehicle hardware.
- ❖ **Clear first users**
University autonomy labs + AV R&D teams can compare planners before physical trials.
- ❖ **Scale by validation depth**
Synthetic → 2 scenes → sensor-fault stress → HIL / physical trials.

Known risks → Engineering Response

- ❖ **Dense-scene latency**
Median run-level P95 = 130.1 ms; target is <150 ms.
- ❖ **Rare road-user perception**
Cattle, pushcarts and auto-rickshaws are the classes most likely to expose generic-model gaps.
- ❖ **Stop / resume oscillation**
Dense-market and yielding logic can deadlock or chatter even without a collision.

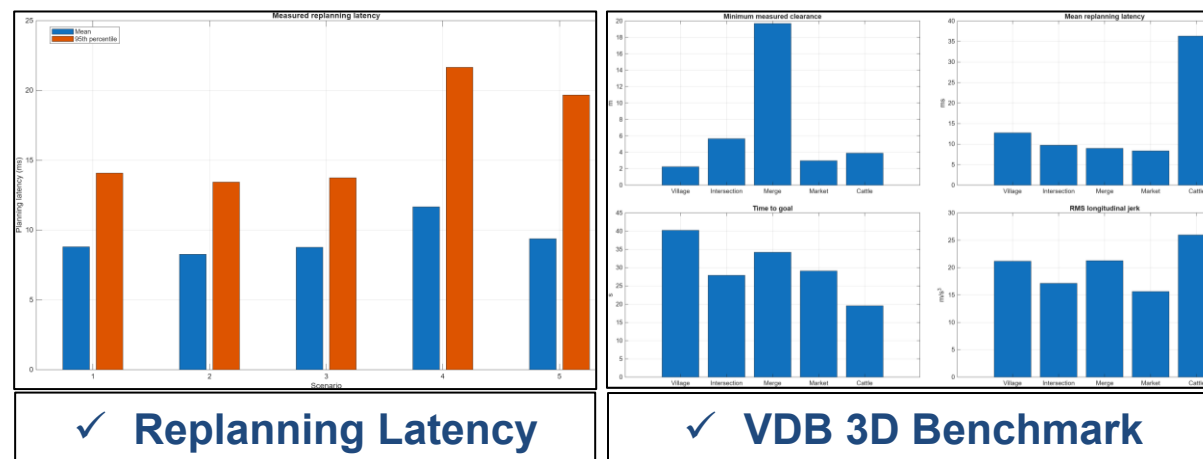
➤ VALIDATION COVERAGE — ACTUAL RUNS

$$5 \text{ Scenarios} \times 12 \text{ Runs Per Scenario} = 60 \text{ Total Tests}$$



- ❖ 60/60 executed
 - ❖ 58 goals
 - ❖ 0 collision episodes
 - ❖ 2 timeouts
- Reached goal
● Timeout

Scenario	Completion	Collision Episodes	Percentile 95 (sec)	Max Jerk (m/s ³)
Unmarked village road	11/12 (91.7%)	Nil	0.1368 s	623.59 m/s ³
Unsignalized urban intersection	12/12 (100.0%)	Nil	0.1301 s	629.22 m/s ³
Arterial slow-traffic merge	11/12 (91.7%)	Nil	0.1443 s	620.31 m/s ³
Dense mixed market	12/12 (100.0%)	Nil	0.1259 s	591.49 m/s ³
Sudden cattle crossing	12/12 (100.0%)	Nil	0.1108 s	658.21 m/s ³



➤ Impact on Target Users

❖ AV / ADAS R&D

India-specific benchmark for behaviour + motion planning in dense, mixed traffic.

❖ Safety / Validation

Audit every run: collision, clearance, timing, comfort and completion - not only a demo video.

❖ Universities / Labs

Re-playable environment to compare planners, reproduce failures and teach closed-loop autonomy

➤ Benefits of the solution

❖ Social / Safety

Tests informal merges, sudden crossings and mixed road users; falls back conservatively when risk rises.

❖ Economic / R&D

Reusable virtual edge cases reduce repeated physical trials and shorten debug cycles.

❖ Environmental

Virtual-first validation reduces early prototype kilometres and test-resource use

❖ Technical / Trust

Predictions → behaviour → trajectory
→ safety status → metrics are re-playable and auditable.

One Fixed Prediction

Assumes a nearby road user will continue in only one expected way

Lane-dependent Planning

Works best when lane markings and road structure are clear and predictable.

One Successful Demo

A single simulation video can't show whether the system is consistently reliable.

With Our Approach

Possible Road-user Actions

Considers actions like crossing, stopping, turning before choosing a safe path.

Road-space + Traffic-aware

Uses drivable space and road users, so decisions work when lane marks are weak

Repeatable, Measurable Tests

Logs collisions, clearance, replanning time and completion for replay and checks.

➤ POC Validation Targets on Test Machine

✓ **0 Collisions**

Nominal Completed Runs

✓ **< 150 ms p95 replanning**

Local Planning Loop

✓ **≥ 95% Completion**

Across Required Scenarios

✓ **Jerk + Curvature**

Than Baseline

5 Required Scenarios

1 Unmarked village

2 Unsignalized junction

3 Highway merge

4 Dense market

5 Cattle crossing

➤ Indian-road Data


[Click Here](#)

➤ India Driving Dataset (IDD) | IIIT Hyderabad / INSAAN

IDD directly matches this problem, offering 10,000 Indian road images, 34 classes and 182 sequences, capturing complex, weakly structured traffic beyond Western lane-centric benchmarks.


[Click Here](#)

➤ Critical Unstructured Road Scenarios | IEEE Intelligent Vehicles 2025

IDD-CRS focuses on critical driving events, with 90 hours of Hyderabad data, 5,400 videos and 135,000 frames, enabling safety-focused validation of risky interactions, hazardous situations and hard negatives.


[Click Here](#)

➤ Indian Object-Detection Dataset for Unstructured Traffic | Data in Brief 2022

DATS_2022 offers 10,000+ Indian traffic images across 45 classes, covering rural and urban road users such as cattle, animal carts, rickshaws and other uncommon objects for safer, robust re-planning.

➤ Perception & Fusion


[Click Here](#)

➤ MathWorks Tracking and Sensor Fusion | MathWorks R2026b documentation

This MATLAB/Simulink fusion reference covers radar-camera tracking, Kalman filtering and motion models, converting asynchronous detections into stable tracks with position, velocity and uncertainty for SIH.


[Click Here](#)

➤ Center-Based Radar-Camera Fusion for 3D Detection | Nabati & Qi | 2020/2021 research

Center-Fusion combines camera semantics with radar range and velocity using frustum-based association, improving depth, rotation and velocity estimation and reporting over 12% higher nu-Scenes Detection Score.


[Click Here](#)

➤ Unified Bird's-Eye-View Multi-Sensor Fusion | Liu et al. | 2022 / ICRA 2023 lineage

BEVFusion combines camera semantics and LiDAR geometry in a shared bird's-eye-view, providing an efficient perception interface for path planning where lanes are absent or unreliable.

➤ Motion Prediction


[Click Here](#)

➤ Pedestrian Intent & Trajectory in Unstructured Traffic | Bokkasam et al. | ICRA 2025

IDD-Pedestrian targets pedestrian intention and trajectory prediction in unstructured Indian traffic, covering occlusion, poor illumination and unsignalized interactions, while exposing weaknesses in conventional pedestrian prediction methods.


[Click Here](#)

➤ Vehicle-Following in Mixed Indian Traffic | Kashyap | Transport Research Record 2020

This study examines mixed Indian traffic without strict lane discipline, modeling strict and staggered following to capture lateral and longitudinal interactions missed by purely map-based forecasting.


[Click Here](#)

➤ A Unified Architecture for Predicting Multiple Agent Trajectories | ICLR 2022

Scene Transformer predicts consistent future trajectories for multiple interacting road users simultaneously, capturing how vehicles, pedestrians and the autonomous vehicle influence each other in dynamic driving scenes.

➤ Planning & Validation


[Click Here](#)

➤ Automated Driving Toolbox - ADAS/Autonomous Workflow | MathWorks R2026b

Automated Driving Toolbox integrates perception, scenarios, sensor models, vehicle interfaces and validation, supporting MIL, SIL and HIL workflows beyond basic scripted autonomous-driving simulations.


[Click Here](#)

➤ ASAM OpenSCENARIO 2.0 - Dynamic Scenario Description | ASAM Standard

ASAM OpenSCENARIO standardizes dynamic traffic scenarios, interactions and parameterization, enabling RoadRunner-exported tests to be portable, reusable and systematically varied instead of one-off demonstrations.


[Click Here](#)

➤ Hybrid A* Path Planner for Nonholonomic Vehicles | MathWorks Navigation Toolbox

Hybrid A* searches x-y-heading states while respecting vehicle motion, using Reed-Shepp expansion for efficient planning on unmarked roads and around static obstacles, potholes, blocked paths and complex open spaces.

Our solution perfectly aligns with the Sustainable Development Goals.

